Aneesh Rangnekar

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Research Summary

Senior AI Scientist in Computational Radiology at AstraZeneca, responsible for developing radiology imaging biomarkers from CT, MRI, and PET to inform indication selection, early assessment of biological activity, and patient stratification in oncology. Research focuses on making vision and multimodal foundation models reliable for clinical use: robustness to distribution shift, uncertainty quantification, and out-of-distribution detection. Previously developed and deployed self-supervised and multimodal foundation models for radiology across multi-hospital clinical workflows at Memorial Sloan Kettering.

Research Interests

Imaging biomarkers and radiomics; foundation models for medical imaging; robustness under distribution shift; uncertainty quantification and out-of-distribution detection; multimodal and vision-language models; trustworthy deployment of AI in healthcare.

Technical Skills

Modeling: Foundation models (ViTs, SAM, multimodal VLMs), radiomics, self-supervised learning (DINO, MAE, SimCLR), representation learning, OOD detection, uncertainty estimation, robustness
Systems: Multi-GPU distributed training (Slurm), end-to-end training and evaluation pipelines, data-scaling studies
Tools: PyTorch, JAX, Hugging Face, MONAI, Docker, Slurm, Weights & Biases
Languages: Python, C++, MATLAB

Selected Publications

A Rangnekar, H Veeraraghavan. *Tumor-anchored deep feature random forests for out-of-distribution detection in lung cancer segmentation.* Transactions on Machine Learning Research, 2026.

A Rangnekar, J Miranda, N Horvat, et al. *Tumor-aware augmentation with task-guided attention analysis improves rectal cancer segmentation from magnetic resonance images.* Biomedical Signal Processing and Control, 2026.

A Rangnekar, J Tapias Gomez, J Deasy, H Veeraraghavan. *Parameter-efficient pretrained-CT-to-MRI transfer for rectal cancer segmentation: Performance–calibration trade-offs.* MICCAI Workshop on Medical Image AI in Radiation Therapy (MIART), 2026.

A Rangnekar, N Mankuzhy, J Willmann, CMS Choi, A Wu, M Thor, A Rimner, H Veeraraghavan. *Transformer-based cardiac substructure segmentation from contrast and non-contrast computed tomography for radiotherapy planning.* Physics and Imaging in Radiation Oncology, 2026.

A Rangnekar, H Veeraraghavan. *Random forest-based out-of-distribution detection for robust lung cancer segmentation.* SPIE Medical Imaging, 2026.

J Tapias Gomez, D Kanata, **A Rangnekar**, C Lee, J Garcia-Aguilar, JJ Smith, H Veeraraghavan. *Dual cross-attention siamese transformer for rectal tumor regrowth assessment in watch-and-wait endoscopy.* ISBI, 2026.

M Kayser, M Gridnev, W Wang, M Bain, **A Rangnekar**, A Chatterjee, A Petrov, H Veeraraghavan, N Swinburne. *brat: Aligned multi-view embeddings for brain MRI analysis.* WACV, 2026.

CMS Choi, **A Rangnekar**, J Jiang, H Veeraraghavan. *Topological data analysis visualization for interpretable assessment of AI contouring quality.* SPIE Medical Imaging, 2026.

J Simeth, **A Rangnekar**, V Brennan, S McBride, N Tyagi, H Veeraraghavan. *Out-of-domain data generalization using parallel-route coherent manifold-mixup for prostate cancer segmentation.* SPIE Medical Imaging, 2026.

A Rangnekar, N Nadkarni, J Jiang, H Veeraraghavan. *Quantifying uncertainty in lung cancer segmentation with foundation models applied to mixed-domain datasets.* SPIE Medical Imaging, 2025.

J Jiang, **A Rangnekar**, H Veeraraghavan. *Self-supervised learning improves robustness of deep learning lung tumor segmentation models to CT imaging differences.* Medical Physics, 2025.

J Jiang, **A Rangnekar**, H Veeraraghavan. *Co-distilled attention guided masked image modeling with noisy teacher for self-supervised learning on medical images.* MIDL, 2025.

A Rangnekar, C Kanan, M Hoffman. *Semantic segmentation with active semi-supervised learning.* WACV, 2023.

A Rangnekar, C Kanan, M Hoffman. *Semantic segmentation with active semi-supervised representation learning.* BMVC, 2022.

A Rangnekar, N Mokashi, E Ientilucci, C Kanan, M Hoffman. *AeroRIT: A new scene for hyperspectral image analysis.* IEEE TGRS, 2020.

B Uzkent, **A Rangnekar**, M Hoffman. *Tracking in aerial hyperspectral videos using deep kernelized correlation filters.* IEEE TGRS, 2018.

B Uzkent, **A Rangnekar**, M Hoffman. *Aerial vehicle tracking by adaptive fusion of hyperspectral likelihood maps.* CVPR Workshops, 2017.

Preprints

S Zaman, **A Rangnekar**, N Lee, H Veeraraghavan. *Adaptive computation with conformal certification.* Under review; arXiv:2610.19189.

J Tapias Gomez, D Kanata, **A Rangnekar**, et al. *Prediction of rectal cancer regrowth from longitudinal endoscopy.* Under review; arXiv:2605.12855.

CMS Choi, **A Rangnekar**, J Jiang, GD Hugo, MR Waters, N Tyagi, M Hyun, JM Moran, H Veeraraghavan. *TopoVis: Topological data analysis-based segmentation visualization for interpretable assessment of AI contouring quality.* Under review.

Selected Research Projects

Self-supervised 3D foundation models for medical imaging

Representation learning and scaling, Memorial Sloan Kettering

- Built self-supervised pretraining pipelines for 3D CT volumes with transformer backbones (ViT3D, Swin3D, and Hiera3D), adapting iBOT-style masked patch distillation and DINOv2-style objectives to volumetric data.
- Compared hierarchical and standard ViT backbones within a unified training framework to study representation structure and generalization.
- Studied how pretraining dataset size affects representation quality, toward scaling laws for 3D medical scans.

Multimodal medical foundation models for robustness and OOD generalization

Multimodal learning, language supervision, and model reliability, Memorial Sloan Kettering

- Extended self-supervised pipelines from CT to multimodal CT, MRI, and PET, adding language supervision to align imaging representations with clinical concepts.
- Benchmarked robustness under modality shift and missing modalities across scanners and clinical settings.
- Evaluated whether language-informed representations improve interpretability and out-of-distribution detection in safety-critical applications.

Experience

AstraZeneca Aug. 2026 – Present

Senior AI Scientist, Computational Radiology *Boston, MA*

- Responsible for developing radiology imaging biomarkers in the Radiology & Theranostics team (Cancer Biomarker Development), using radiomics and deep learning on CT, MRI, and PET.
- Building imaging-derived analytics that inform indication selection, early assessment of biological activity, and patient stratification across oncology programs.

Memorial Sloan Kettering Cancer Center Aug. 2022 – Aug. 2026

Research Fellow (AI for Medical Imaging) *New York, NY*

- Developed and scaled foundation model training pipelines across multi-institutional datasets (CT, MRI, PET), enabling robust representation learning under real-world distribution shifts.
- Applied parameter-efficient fine-tuning (LoRA/DoRA) to pretrained imaging encoders (∼100M parameters), reducing training compute by ∼30% and trainable parameters by ∼90% while improving generalization on unseen clinical distributions.
- Designed end-to-end evaluation pipelines for oncology imaging models, including failure-mode analysis and robustness assessment under scanner, protocol, and population shifts.
- Developed out-of-distribution detection pipelines for safety-critical deployment, enabling reliable monitoring of model performance across heterogeneous multi-hospital data streams.
- Collaborated with clinicians and domain experts to translate real-world scientific problems into machine learning objectives, training signals, and evaluation criteria.
- Mentored students and research collaborators on medical imaging and machine learning projects.

Rochester Institute of Technology May 2016 – Aug. 2022

Graduate Research Assistant *Rochester, NY*

- Developed self-supervised and semi-supervised representation learning methods for high-dimensional hyperspectral imagery, achieving >10% improvement over supervised baselines in low-label regimes.
- Released AeroRIT and RooftopHSI, large-scale (4TB+) hyperspectral datasets with benchmarking infrastructure, enabling reproducible research and adoption across the remote sensing community.

SRI International Jun. 2018 – Nov. 2018

Research Intern *Princeton, NJ*

- Designed continual self-supervised approaches for aerial object counting with efficient labeling.
- Adapted Faster R-CNN for occlusion-aware infrared object detection under low-data conditions.

Conduent Labs Jun. 2017 – Aug. 2017

Research Intern *Rochester, NY*

- Developed GAN-based data augmentation for vehicle occupancy estimation at toll booths.

Education

Rochester Institute of Technology 2015 – 2022

Ph.D., Imaging Science *Rochester, NY*

- Research: self-supervised representation learning for high-dimensional hyperspectral data.

Rochester Institute of Technology 2013 – 2015 (Degree: 2017)

M.S., Electrical Engineering *Rochester, NY*

- Research: parameter-efficient image processing for HP scanners.

Service

Reviewer: CVPR, ICCV, ECCV, MICCAI, Medical Physics, IEEE TIP, IEEE TGRS.